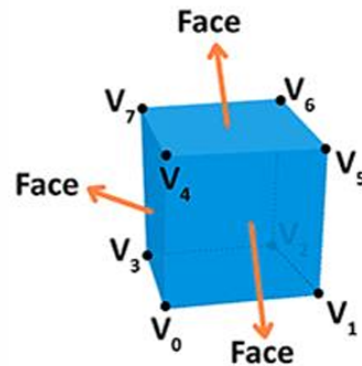


- A **Vertex** is a point in space having its own 3D position in the coordinate system
- **Position**: Identifies it in a 3D space (x, y, z).
- **Color**: Holds an RGBA value (R, G and B for the red, green, and blue channels, alpha for transparency — all values range from 0.0 to 1.0).
- **Normal**: A way to describe the direction the vertex is facing.
- **Texture**: A 2D image that the vertex can use to decorate the surface it is part of instead of a simple color.

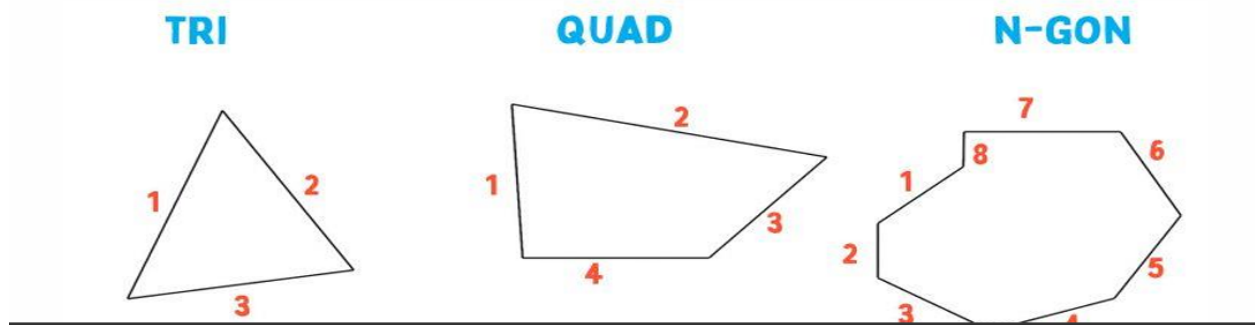


Mesh

- A **mesh** is a **collection of vertices, edges, and faces** that define the shape of a 3D object. The most popular type of polygon mesh used in computer graphics is a Triangle Mesh. Any object, either a mountain or a game character, can be modelled with Triangle meshes.

Faces: Triangle, Quad, N-Gon

- A face can have a minimum of **3 vertices**. This is then a **triangle**. If you have **four vertices** it is called a **quad**, and if you have more than four vertices, then it is simply a **general polygon**.
- When you have **5 or more edges around** a face, it's referred to as an **N-gon**.



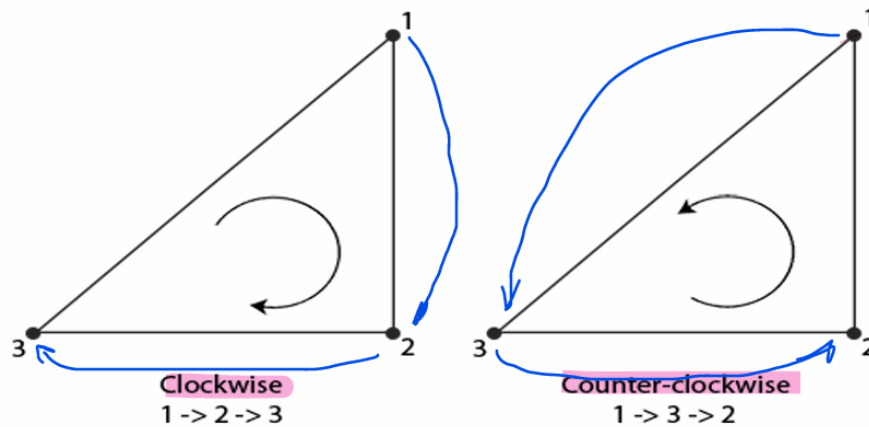
Difficulty when subdividing

- N-Gons **subdivide poorly and produce unwanted topology** **شكل هندسي**, even on flat surfaces. While it may be quicker to model using N-Gons, it is a bad habit to get into as they disrupt the edge flow of your model, making selection of edge loops more difficult.
- Having **N-Gons often leads to strange rendering** or smoothing artifacts that are hard or impossible to eliminate.
- Instead of leaving N-Gons around in your model, it is better to avoid them altogether since quads and tris will always subdivide properly.

Face normals

- A **normal** is just a **hidden vector**, basically a direction without a specified origin.

Winding order



- **FBX** is one of the most used 3d formats.

1. From-Scratch modeling :

In from-scratch modeling, you lay out each vertex and draw each polygon one by one until the entire model is completed

2- Primitive modeling :

start with basic shapes (

- **3-Box Modeling:**
 - Box modeling is a popular approach to character modeling
 - quick creation of basic shapes.
 - start with a **cube** and **extrude** arms, fingers, legs, toes, and a head.
 - You then add **detail** to the whole shape by splitting and refining the model
- **4-Boolean modeling (Constructive Solid Geometry):**
 - additive or subtractive method of changing the geometry of one object by introducing another object and running a Boolean function:
 - Union
 - Intersection
 - Difference

- **5-Laser Scanning**

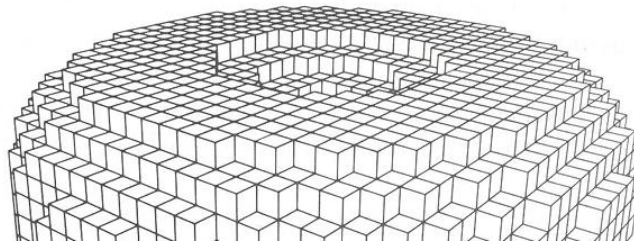
- starts with a real object that you digitally laser-scan to create the geometric surfaces.
- The scanning process can be a quick and easy task,
- but the geometry created is always very Complex

- **6- Digital Sculpting:**

- artists to use a massive polygon resolution to create a surface that can be pushed and pulled like clay صلصال

8-Voxels

- Partition space into uniform grid
 - Grid cells are called a voxels (like pixels)
- Store properties of solid object with each voxel
 - Occupancy
 - Color
 - Density
 - etc.



- **$O(n^3)$ storage for $N \times N \times N$ grid**

9-NURBS modeling

- NURBS stands for **Non-Uniform Rational B-spline**.
- In NURBS modelling, lines and surfaces are **not manipulated by moving vertices**, edges, faces or polygons.
- Instead NURBS surfaces and lines are manipulated by **special control points**.
The following 2 techniques make use of NURBS.

A **NURBS model** is a **mathematical modeling type** commonly used to generate **curves**

1. **Perspective View:** the configuration includes field of view, aspect ratio and optional near and far planes.
A perspective camera is how we see the real world. If we take a look at the things around us, they have depth and we can judge their distance.
2. **Orthographic view:** Objects are drawn without perspective distortion.

View Frustum

الشكل المخروطي للمشاهدة

- The **view frustum** is the **volume** that contains everything that is **potentially** (there may be occlusions) **visible** on the screen.
- This volume is defined according to the camera's settings, and when using a perspective projection takes the shape of a truncated **pyramid**.

What is Z-Fighting?

- **Z-Fighting** is where multiple objects fight to be rendered closest to the camera (*fighting to be on top*). This can happen when the objects are overlapping due to having similar z-buffer values as the scene is rendered.
- **Textures** are 2D images used in the 3D space to make the objects look better and more realistic. Textures are

Types of texture Maps-p1

1. **Color map, diffuse** map, or **Albedo** map: Gives the object **colours**
2. **Metalness** map: Tells you the **metallic** معدني properties of the material.
3. **Roughness** map or **gloss** map: Tells you the **glossiness** لمعان of the surface, determines how **reflective** the material is.
4. **Transparency Map**: grey scale textures that use black and white values to signify areas of transparency or opacity on an object's material

Gloss (or Roughness)

(أو خشونة) لمعان

- The terms **Gloss** and **Roughness** are interchangeable - they are simply the **inverts** of each other. كلمتان متعاكستان
- So if your software calls it "Gloss" like (Marmoset), but there is only a "Roughness" map provided, simply invert it in the software (usually a checkbox is provided) before connecting.
- It often adds a lot of character to hard surface materials like wood or metal

Types of texture Maps-p2

5. **Bump map:** Creates the illusion of depth by faking **اصطناع بدون تغيير فعلي** details on the surface using height information to tell when a point should be up or down. The surface however is still flat and not actually warped.

Types of texture Maps-p3

- 6- **Normal map:** A type of bump map that fakes the details by using angle information to tell which direction each point of the surface should be oriented towards. This gives better effect with lighting and shadows.

Types of texture Maps-p4

- 7- **Displacement map:** Similar to a height map but actually changes the geometry of the object by moving vertices (points) on the mesh (surface). This physically alters the form of the object.

Normal Maps VS. Bump Maps

- both affect the normals of your geometry and create the illusion of detail without having to rely on extra geometry
- **bump maps** just encode height information using black and white values
- **Black:** minimum height delta
- **White:** maximum height delta
- **normal maps** use RGB values to signify the orientation of the surface normals:
- red, green and blue channels = X, Y and Z orientation of the surface
- Normal maps can typically get **more detailed** information onto the surface

ممکن یچی صح و غلط رکز

PBR Textures

- **PBR**, or **Physically Based Rendering**, is a model of shading used in computer graphics that more closely mimics the flow of light in the real world.
- PBR textures produce a higher quality of realness with greater variations of lighting

PBR Textures Components

To create material in specular workflow you will need the following maps:

1. **Albedo (Diffuse) Map** – base color input, which is the most important map, giving information about the color of the object.
2. **Specular Map** – defines an object's shininess by using a greyscale map. In its practical application the brighter the shade, the more light will be reflected.
3. **Glossiness Map** – this map controls the reflectivity of an object, whether the reflection of light will be blurred or sharp.
4. **Normal Map** – this map determines the number of dents and bumps an object has. Similar to a bump map yet more advanced, the normal map is saved in RGB format, containing its information in XYZ axes.
5. **Height (Displacement) Map** – this map is saved as a grayscale which creates a larger group of convexities and creases which affect the objects mesh.

<https://cgaxis.com/how-to-use-pbr-textures/>

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Note : U and V values range from 0 to 1 مهم جدا جدا جدا

Rendering types

1-Clay Model Rendering::

- When 3D modeler's present their modelling skills they often produce a clay render. Clay render means a rendering which looks like a picture of a clay model.
- This is not really a rendered effect however as it is generally the default form of a non textured model
- Useful for design

Rendering types

2-Wireframe Rendering::

The purpose of wire rendering is to reveal the polygon structure of the model. Wire rendering displays only edges of the polygons. Wire render and clay render are often combined.

Rendering types

3-Cell Shading Rendering::

Cell Shading is a form of **Non-photo realistic rendering (NPR)**. In contrast to traditional computer graphics, which has focused on photo realism, NPR is inspired by artistic styles such as painting, drawing, technical illustration, and animated cartoons

4-Photorealistic Rendering

- **Photorealistic Rendering** is a modern technique that renders the images more realistic and clearer than ever.

3D animation Phases

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ای هی خطوات
3d animation

1- Modelling the character:

to create the basic model

2- Rigging the character:

to add bones

3- Skinning the character:

to make a relation between model parts and
bones

4- Animating the character

Character rigging means a process of creating the bone structure for a character.